
Table of Contents

EAS 4300 HW #8 Question 1	1
Given Parameters	1
Inlet, Compressor, Fan, and Burner	2
Turbine, Nozzles, and Performance for Each Bypass Ratio	3
Optimum Bypass Ratio	4
Figure 1: Specific Thrust vs. Bypass Ratio	5
Figure 2: TSFC vs. Bypass Ratio	5
Figure 3: Efficiencies vs. Bypass Ratio	6
Figure 4: Exit Mach Numbers vs. Bypass Ratio	7
Save Data to CSV	8

EAS 4300 HW #8 Question 1

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You are to specify the bypass ratio of a turbofan engine for a commercial aircraft that is to cruise at a Mach number of 0.8. The exhaust from two streams are not internally mixed and both streams are ideally expanded. The design altitude is 10,000 m. The fuel has a heat of combustion of 43,000 kJ/kg, and a stoichiometric fuel-to-air ratio of 0.06. Assume the gamma is constant at 1.4 throughout the fan, 1.4 in the core up to the entrance of the burner, and 1.35 for the rest of the core (burner, turbine, and core nozzle). Assume the gas constant is fixed at $R = 287$ J/kgK. The maximum temperature at the inlet of the turbine is 1500 K. The compressor pressure ratio is 24, and the fan pressure ratio is 2.0. Plot I, TSFC, η_a , η_p , η_{th} vs. bypass ratio. Is there an optimum bypass ratio that maximizes the overall efficiency? Use the following efficiencies: $\eta_d = 0.94$, $\eta_c = 0.87$, $\eta_f = 0.92$, $\eta_b = 0.98$, $\eta_{t1} = 0.97$, $\eta_{t2} = 0.85$, $\eta_{cn} = 0.97$, $\eta_{fn} = 0.98$.

```
clc; clear; close all;
```

Given Parameters

```
M_f = 0.8; % Flight Mach number
Ta = 223.252; % Ambient temperature at 10 km [K]
Pa = 26500; % Ambient pressure at 10 km [Pa]
R = 0.287; % Gas constant [kJ/(kg*K)]
T04_max = 1500; % Maximum turbine inlet temperature [K]
dhc = 43000; % Fuel heat of combustion [kJ/kg]
f_stoic = 0.06; % Stoichiometric fuel-to-air ratio

% Pressure ratios
rc = 24; % Compressor pressure ratio
rb = 0.97; % Burner pressure loss factor
rf = 2; % Fan pressure ratio

% Efficiencies
eta_d = 0.94; % Diffuser efficiency
eta_c = 0.87; % Compressor efficiency
eta_b = 0.98; % Burner efficiency
```

```

eta_t = 0.85; % Turbine efficiency
eta_f = 0.92; % Fan efficiency
eta_cn = 0.97; % Core nozzle efficiency
eta_fn = 0.98; % Fan nozzle efficiency

% Specific heat ratios
% k(1): inlet, diffuser, compressor, fan
% k(2): burner, turbine, core nozzle
k = [1.4 1.35];
Cp = (k ./ (k - 1)) * R; % Specific heats [kJ/(kg*K)]

% Bypass ratio range. Above beta = 7.26 the turbine can no longer drive
% the fan and still leave enough pressure to expand the core flow to Pa.
beta = linspace(2, 7.26, 100)';
n = length(beta);

```

Inlet, Compressor, Fan, and Burner

These do not depend on the bypass ratio.

Inlet and diffuser (adiabatic, so $T_{02} = T_{0a}$):

$$T_{0a} = T_a \left(1 + \frac{\gamma_1 - 1}{2} M^2 \right), \quad u = M \sqrt{\gamma_1 R T_a}$$

$$T_{02s} = T_a + \eta_d (T_{0a} - T_a), \quad P_{02} = P_a \left(\frac{T_{02s}}{T_a} \right)^{\gamma_1 / (\gamma_1 - 1)}$$

Compressor and fan:

$$T_{03} = T_{02} + \frac{T_{02} r_c^{(\gamma_1 - 1) / \gamma_1} - T_{02}}{\eta_c}, \quad w_c = c_{p1} (T_{03} - T_{02}), \quad P_{03} = r_c P_{02}$$

$$T_{07} = T_{02} + \frac{T_{02} r_f^{(\gamma_1 - 1) / \gamma_1} - T_{02}}{\eta_f}, \quad P_{07} = r_f P_{02}$$

Burner energy balance, with c_{p1} for the incoming air and c_{p2} for the products:

$$c_{p1} T_{03} + f \eta_b \Delta h_c = (1 + f) c_{p2} T_{04} \Rightarrow f = \frac{c_{p2} T_{04} - c_{p1} T_{03}}{\eta_b \Delta h_c - c_{p2} T_{04}}, \quad P_{04} = r_b P_{03}$$

If f would exceed f_{st} , the burner runs at f_{st} and T_{04} is solved from the same balance.

```

% Inlet (diffuser) conditions
T0a = Ta * (1 + ((k(1)-1)/2)*M_f^2);
T02 = T0a;
T02s = eta_d * (T02 - Ta) + Ta;
P02 = Pa * (T02s/Ta)^(k(1)/(k(1)-1));
U = M_f * sqrt(k(1)*R*1000*Ta); % Freestream velocity [m/s]

% Compressor

```

```

T03s = T02 * rc^((k(1)-1)/k(1));
T03 = (T03s - T02)/eta_c + T02;
Wc = Cp(1) * (T03 - T02); % Compressor work per kg of core air
[kJ/kg]
P03 = rc * P02;

% Fan
T07s = T02 * rf^((k(1)-1)/k(1));
T07 = (T07s - T02)/eta_f + T02;
P07 = rf * P02;

% Burner
Fb = (Cp(2)*T04_max - Cp(1)*T03) / (eta_b*dhc - Cp(2)*T04_max);
T04 = T04_max;
if Fb > f_stoic
    Fb = f_stoic;
    T04 = (Cp(1)*T03 + f_stoic*eta_b*dhc) / ((1 + f_stoic)*Cp(2));
end
P04 = rb * P03;

```

Turbine, Nozzles, and Performance for Each Bypass Ratio

Turbine drives the compressor and the fan (β kg of bypass air per kg of core air):

$$(1 + f) c_{p2}(T_{04} - T_{05}) = w_c + \beta c_{p1}(T_{07} - T_{02}), \quad T_{05s} = T_{04} - \frac{T_{04} - T_{05}}{\eta_t}, \quad P_{05} = P_{04} \left(\frac{T_{05s}}{T_{04}} \right)^{\gamma_2/(\gamma_2-1)}$$

Core nozzle (hot) and **fan nozzle** (cold), both ideally expanded to P_a . For each, with inlet state (T_0, P_0) , efficiency η_n and γ :

$$T_{es} = T_0 \left(\frac{P_a}{P_0} \right)^{(\gamma-1)/\gamma}, \quad T_e = T_0 - \eta_n(T_0 - T_{es}), \quad M_e = \sqrt{\frac{2}{\gamma-1} \left(\frac{T_0}{T_e} - 1 \right)}, \quad u_e = M_e \sqrt{\gamma R T_e}$$

using $(T_{05}, P_{05}, \eta_{cn}, \gamma_2)$ for the core and $(T_{07}, P_{07}, \eta_{fn}, \gamma_1)$ for the fan.

Performance (per kg of core air):

$$I = (1 + f) u_{e,hot} - u + \beta (u_{e,cold} - u), \quad \text{TSFC} = \frac{f}{I}$$

$$\eta_{th} = \frac{(1 + f) u_{e,hot}^2 + \beta u_{e,cold}^2 - (1 + \beta) u^2}{2f \Delta h_c}, \quad \eta_p = \frac{2 I u}{(1 + f) u_{e,hot}^2 + \beta u_{e,cold}^2 - (1 + \beta) u^2}, \quad \eta_0 = \eta_{th} \eta_p$$

```

Wf = zeros(n, 1);
T05 = zeros(n, 1);
P05 = zeros(n, 1);
Me = zeros(n, 1);
M8 = zeros(n, 1);
Ue_hot = zeros(n, 1);

```

```

Ue_cold = zeros(n,1);
I = zeros(n,1);
TSFC = zeros(n,1);
eta_th = zeros(n,1);
eta_p = zeros(n,1);
eta_0 = zeros(n,1);

% Fan nozzle (same for every beta)
T8s = T07 / (P07/Pa)^((k(1)-1)/k(1));
T8 = T07 - eta_fn * (T07 - T8s);
M8(:) = sqrt((2*(T07/T8 - 1))/(k(1)-1));
Ue_cold(:) = M8(1) * sqrt(k(1)*R*1000*T8);

for i = 1:n
    % Turbine: work extraction drives the compressor and fan
    Wf(i) = beta(i) * Cp(1) * (T07 - T02);
    T05(i) = T04 - (Wc + Wf(i)) / ((1 + Fb) * Cp(2));
    T05s = T04 - (T04 - T05(i)) / eta_t;
    P05(i) = P04 * (T05s/T04)^(k(2)/(k(2)-1));
    if P05(i) <= Pa
        error('At beta = %.2f the turbine exit pressure is below Pa; lower
the beta range.', beta(i));
    end

    % Core nozzle: expansion to ambient
    T6s = T05(i) / (P05(i)/Pa)^((k(2)-1)/k(2));
    T6 = T05(i) - eta_cn * (T05(i) - T6s);
    Me(i) = sqrt((2*(T05(i)/T6 - 1))/(k(2)-1));
    Ue_hot(i) = Me(i) * sqrt(k(2)*R*1000*T6);

    % Specific thrust and TSFC
    I(i) = (1 + Fb) * Ue_hot(i) - U + beta(i)*(Ue_cold(i) - U);
    TSFC(i) = Fb / I(i);

    % Efficiencies
    KE = (1 + Fb)*Ue_hot(i)^2 + beta(i)*Ue_cold(i)^2 - (beta(i)+1)*U^2;
    eta_th(i) = KE / (2*Fb*dhc*1000);
    eta_p(i) = 2*I(i)*U / KE;
    eta_0(i) = eta_th(i) * eta_p(i);
end

```

Optimum Bypass Ratio

```

[eta_0_max, i0] = max(eta_0);
[I_max, iI] = max(I);
[TSFC_min, iT] = min(TSFC);
fprintf('Fuel-to-air ratio: f = %.4f\n', Fb);
fprintf('Maximum overall efficiency: eta_0 = %.4f at beta = %.2f\n',
eta_0_max, beta(i0));
fprintf('Maximum specific thrust: I = %.1f m/s at beta = %.2f\n', I_max,
beta(iI));
fprintf('Minimum TSFC: %.4e s/m at beta = %.2f\n', TSFC_min, beta(iT));

```

Fuel-to-air ratio: $f = 0.0241$
Maximum overall efficiency: $\eta_0 = 0.2812$ at $\beta = 5.88$
Maximum specific thrust: $I = 1218.3$ m/s at $\beta = 5.88$
Minimum TSFC: 1.9818×10^{-5} s/m at $\beta = 5.88$

Figure 1: Specific Thrust vs. Bypass Ratio

```
figure;  
plot(beta, I, 'LineWidth', 1.5)  
hold on  
plot(beta(iI), I_max, 'o', 'MarkerSize', 6)  
hold off  
xlabel('\beta', 'FontWeight', 'bold')  
ylabel('Specific Thrust (m/s)', 'FontWeight', 'bold')  
title('Specific Thrust vs. Bypass Ratio')  
grid on
```

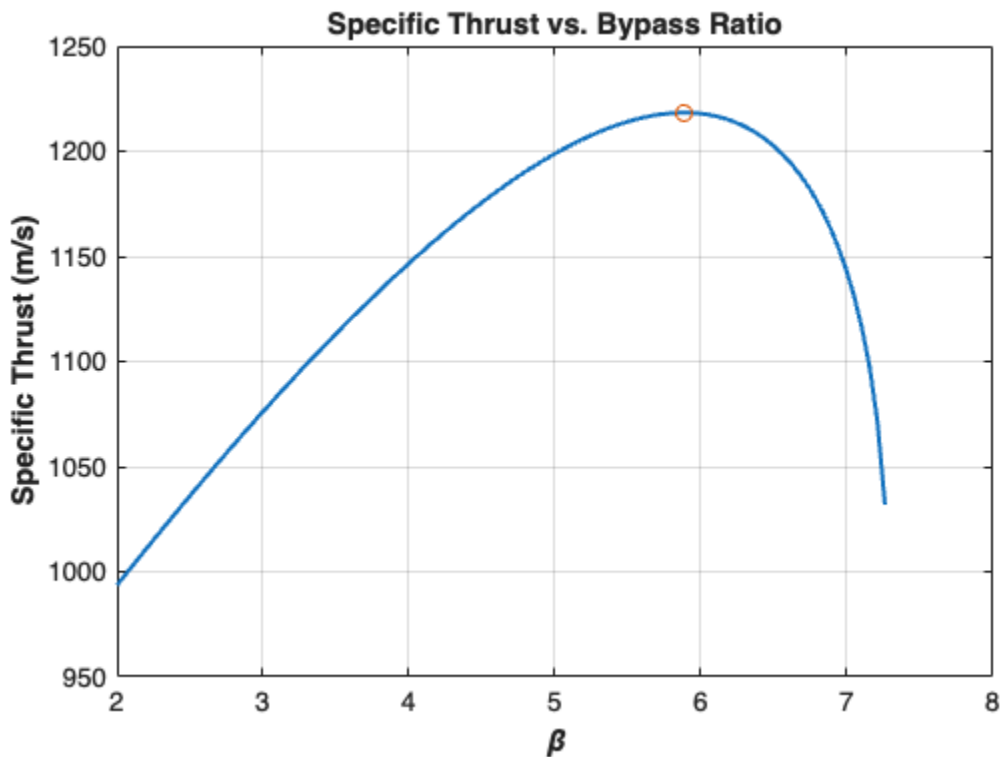


Figure 2: TSFC vs. Bypass Ratio

```
figure;  
plot(beta, TSFC, 'LineWidth', 1.5)  
hold on  
plot(beta(iT), TSFC_min, 'o', 'MarkerSize', 6)  
hold off  
ax = gca;  
ax.YAxis.Exponent = 0;  
xlabel('\beta', 'FontWeight', 'bold')
```

```
ylabel('TSFC (s/m)', 'FontWeight', 'bold')
title('TSFC vs. Bypass Ratio')
grid on
```

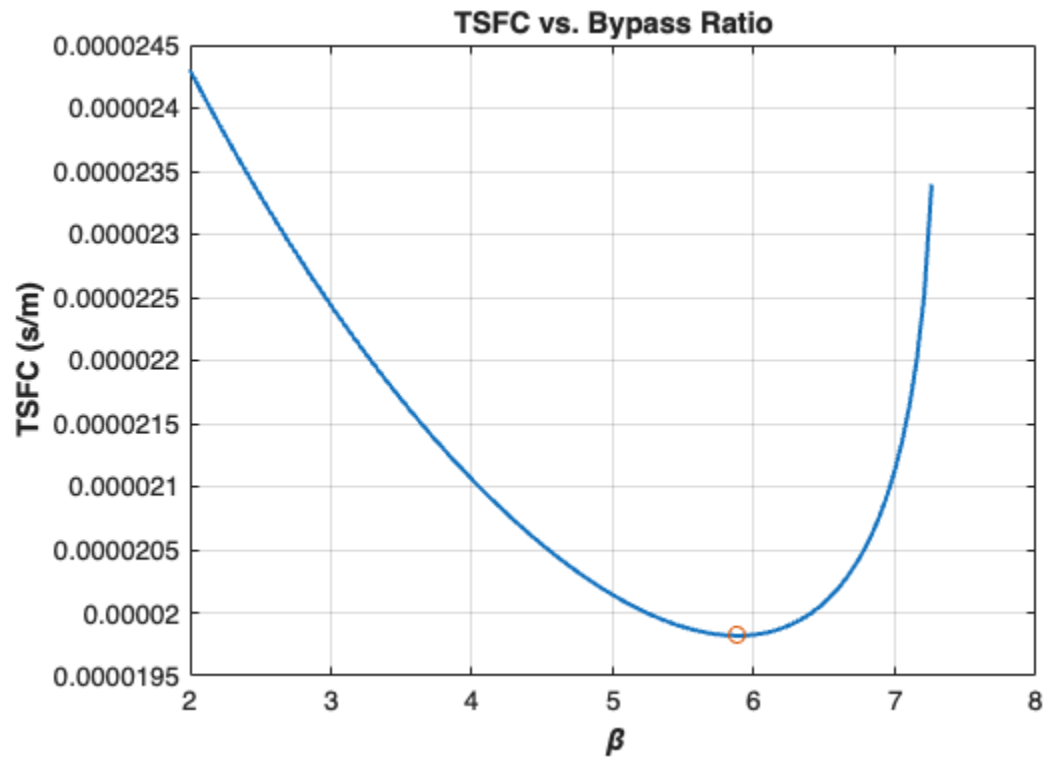


Figure 3: Efficiencies vs. Bypass Ratio

```
figure;
plot(beta, eta_th, 'LineWidth', 1.5)
hold on
plot(beta, eta_p, 'LineWidth', 1.5)
plot(beta, eta_0, 'LineWidth', 1.5)
plot(beta(i0), eta_0_max, 'ko', 'MarkerSize', 6)
text(beta(i0), eta_0_max, sprintf(' max \eta_0 = %.3f at \beta = %.2f',
eta_0_max, beta(i0)), ...
'VerticalAlignment', 'bottom', 'FontSize', 8)
hold off
xlabel('\beta', 'FontWeight', 'bold')
ylabel('Efficiency', 'FontWeight', 'bold')
title('Efficiencies vs. Bypass Ratio')
legend('\eta_{th}', '\eta_{p}', '\eta_{0}', 'Location', 'best')
grid on
```

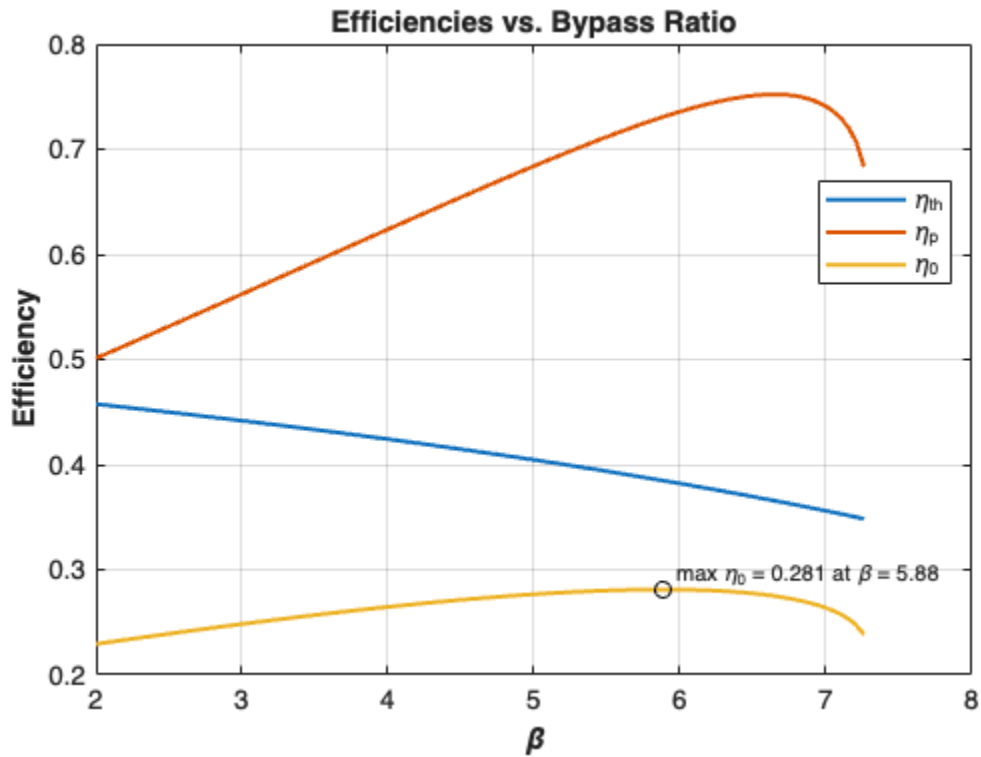
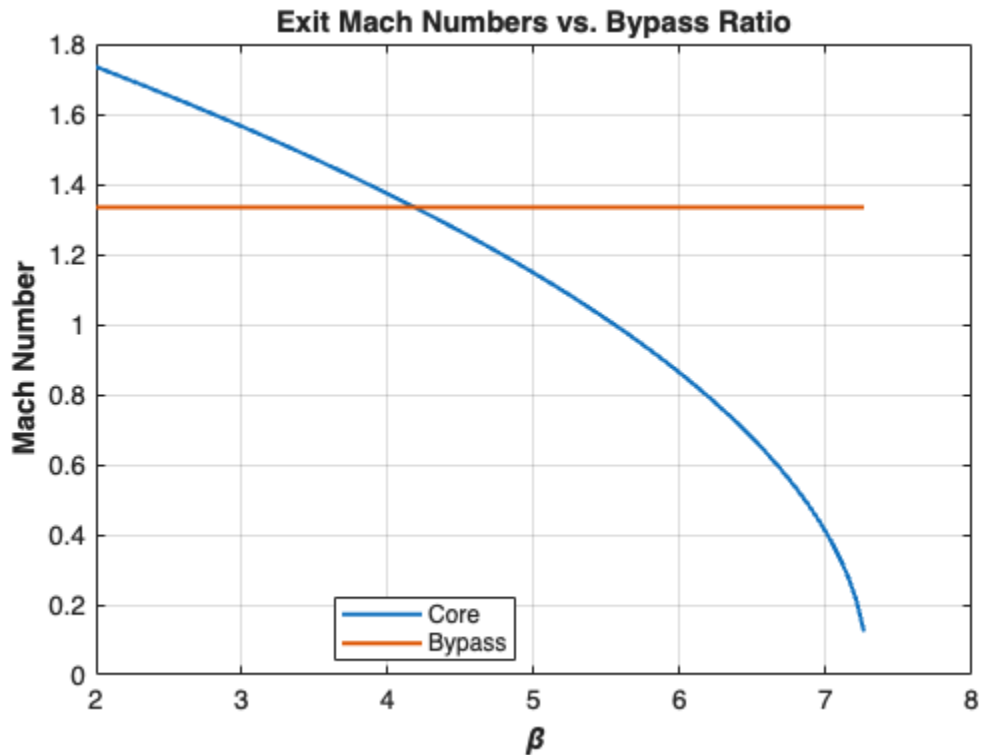


Figure 4: Exit Mach Numbers vs. Bypass Ratio

```
figure;
plot(beta, Me, 'LineWidth', 1.5)
hold on
plot(beta, M8, 'LineWidth', 1.5)
hold off
xlabel('\beta', 'FontWeight', 'bold')
ylabel('Mach Number', 'FontWeight', 'bold')
title('Exit Mach Numbers vs. Bypass Ratio')
legend('Core', 'Bypass', 'Location', 'best')
grid on
```



Save Data to CSV

```
dataTable = table(beta, I, TSFC, eta_th, eta_p, eta_0, Me, M8, repmat(Fb, n,  
1), ...  
    'VariableNames',  
{ 'BypassRatio', 'SpecificThrust', 'TSFC', 'ThermalEff', 'PropEff', ...  
    'OverallEff', 'CoreMach', 'BypassMach', 'FuelAirRatio' });  
writetable(dataTable, 'HW8_Q1_data.csv');  
disp('Data saved to HW8_Q1_data.csv');
```

Data saved to HW8_Q1_data.csv

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